

Polymorphism and Encapsulation

[]: Magic Methods

```
[9]: class Point:
      def __init__(self, x, y):
          self.x = x
          self.y = y

      def __repr__(self):
          return f"Point({self.x}, {self.y})"

p1 = Point(2, 3)
print(p1)

Point(2, 3)
```

[]: Inheritance

```
[7]: class Animal:
      def __init__(self, name):
          self.name = name

      def speak(self):
          print(self.name)

class Dog(Animal):
    pass

d1 = Dog("visii")
d1.speak()

visii
```

[]: Polymorphism

```
[8]: class Cat:
      def sound(self):
          print("HI BRO")

class Fox:
    def sound(self):
        print("MY NAME IS ADITYA THAKUR ")

c1 = Cat()
f1 = Fox()

c1.sound()
f1.sound()

HI BRO
MY NAME IS ADITYA THAKUR
```

[]: Encapsukation

```
[10]: class ScoreBoard:
      def __init__(self, score):
          self.__score = score

      def get_score(self):
          return self.__score

s1 = ScoreBoard(1)
print(s1.get_score())
```

1